Adaptive Context Transformer for Semisupervised Remote Sensing Image Segmentation

Yunbo Li, Zhiyu Yi, Yuebin Wang, *Member, IEEE*, and Liqiang Zhang, *Member, IEEE*

***Abstract*— Current deep learning methods for semantic segmentation in remote sensing heavily depend on a substantial amount of labeled data. However, obtaining pixel-level labeled data in this field is both time-consuming and laborious. To address this challenge, semisupervised learning (SSL) methods have been introduced. Pseudosupervision is one of the most effective methods, which can be adopted to enhance the performance of semisupervised semantic segmentation of remote sensing images. But incorrect pseudolabels can cause substantial distortions to the segmentation model in SSL. Moreover, it is difficult for conventional semantic segmentation methods to deal with global–local features of the remote sensing image without adaptive context feature. In this article, we propose a novel learning approach based on an adaptive context transformer (ACT) and pseudolabeling, called ACT for semisupervised (ACTSS) remote sensing image segmentation. We propose an adaptive context attention model with adjustable sliding windows. A small window is used to capture Query (Q) for local feature, and bigger windows are used to capture Key (K) and Value (V) for global feature. Then, we combine them and get the global–local features. And we propose a point–line–plane (PLP) pseudolabel filter mechanism based on clustering and boundary extraction, which can filter unreliable pseudolabels from three angles: point, line, and plane. To validate the effectiveness of the model, we carried out extensive experiments on the LOVEDA, Potsdam, and Vaihingen datasets and compared ACTSS with other methods. These experiments demonstrate that ACTSS achieves state-of-the-art performance for semisupervised semantic segmentation on all tested datasets.**

***Index Terms*— Adaptive context transformer (ACT), pseudolabel filter, semantic segmentation, semisupervised learning (SSL).**

I. Introduction

With the advancement of sensor technology, massive amounts of fine-resolution remotely sensed images are being collected daily [1]. Semantic segmentation classifies each pixel in an image into a corresponding category to achieve pixel-level classification, which plays a significant role in different application scenarios, such as environmental detection, urban planning, and even disaster assessment [2].

Different from ordinary camera imaging, images taken by airborne or space-borne equipment are typically affected to different degrees by factors such as ground sampling area, atmospheric attenuation, and atmospheric path radiation [3]. So it may be more difficult to segment those images. The traditional methods for semantic segmentation often utilize the low-level features extracted from color, shape, and texture [4], [5]. Then, these features are clustered and classified in high-level space for whole image segmentation. The representative models include the linear classifier model [61], support vector machine model [6], decision tree model [7], naive Bayesian model [8], K-means clustering model [9], and integrated learning model [10]. However, the effectiveness of the model significantly depends on the artificially designed features, which could restrict its ability [11], [12].

Nowadays, deep learning has shown its remarkable progress in remotely sensed semantic segmentation [13]. On the one hand, the structure of networks plays an important role in semantic segmentation. The famous one is convolutional neural networks (CNNs), which can capture deeper features from the images due to the powerful feature representation capability [14]. However, it is difficult for CNN to obtain global features limited by the receptive fields. Transformer [15] can solve the problem partly. Multihead attention is the most important part in the transformer layers, which can obtain global contextual information in the image [16], [17], [18]. On the other hand, a large number of annotated data are needed when model is trained on a fully supervised model, but the pixelwise annotation of remote sensing image is exceedingly expensive [35]. The performance of the model is always limited by the lack of labels. Then, the semisupervised learning (SSL) is proposed to reduce annotation requirements, and it becomes more and more popular [19], [20]. For example, self-training [47] utilized unlabeled data and trained the model on pseudolabels. Then, Ouali et al. [45] proposed a cross-consistency training (CCT) network based on consistency regularization, which achieve great success. Moreover, some scholars introduce generative adversarial networks into SSL, which improves the reliability of the pseudolabel obviously.

However, there are still some limitations for deep learning semantic segmentation and SSL. On the one hand, transformer is always used to replace CNN to capture global features [14],

Manuscript received 5 July 2023; revised 8 August 2023 and 17 September 2023; accepted 21 September 2023. Date of publication 25 September 2023; date of current version 6 October 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 42171388 and Grant 41925006, in part by the Special Fund of Chinese Central Government for Basic Scientific Research Operations in Commonweal Research Institutes under Grant AR2213, and in part by the Industry-Academia Collaboration Program for the Ministry of Education (Piesat Information Technology Company Ltd). *(Corresponding author: Yuebin Wang.)*

Yunbo Li, Zhiyu Yi, and Yuebin Wang are with the School of Land Science and Technology, China University of Geosciences (Beijing), Beijing 100083, China (e-mail: 15871860767@163.com; lightyi@email.cugb.edu.cn; xxgcdxwyb@163.com).

Liqiang Zhang is with the Key Laboratory of Remote Sensing Science, Faculty of Geographical Science, Beijing Normal University, Beijing 100875, China (e-mail: zhanglq@bnu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3318788

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.

[21], but it cannot combine local features and global features together. The reason lies in that the core of the transformer is attention, which requires sliding window of the same size when collecting Query (Q), Key (K), and Value (V). The sizes of the windows cannot be changed flexibly. On the other hand, pseudolabel is important for SSL [22]. However, its reliability is low, which seriously affects the model [23]. Conventional pseudolabel methods only select pseudolabel at the stage of generating a probability map by threshold judgment [24]. The varied levels of semantic features are not taken into account, which results in poor filtering efficiency.

In this article, as shown in Fig. 1, we develop an adaptive context attention semisupervised semantic segmentation method for remotely sensed images, called ACTSS. On the one hand, different from traditional attention, we collect the Query (Q), Key (K), and Value (V) by variable sliding windows. We set a smaller sliding window for Query (Q) to capture the local features, then set bigger sliding windows for Key (K) and Value (V) to capture global features, and combine them together by the attention formula. On the other hand, the low reliability of the pseudolabel affects the model seriously. In this article, we propose a pseudolabel filter (PLP) mechanism based on the three joint views of point, line, and plane based on clustering and boundary extraction, which improves the reliability of pseudolabels. Based on this, the efficiency of semantic segmentation is further improved.

The main contributions include the followings.

1) ACTSS takes the adaptive context transformer (ACT) as the segmentation network and proposes a semisupervised approach, where adaptive context attention is the core of the model. The different sizes of the sliding window are used to collect Q, K, and V. Then, we downsample them by multiscale pooling (MP) and combine the information in different ranges. Finally, we compute the attention to harmonize the coordination between the short- and long-range dependences in both explicit and implicit fusion processes.
2) A pseudolabel filter (PLP) mechanism is proposed based on clustering and boundary extraction. First, we obtain the cluster map and boundary map and then filter out the pseudolabels with too many conflicts. This processing can improve the precision of pseudolabels and alleviate the negative impact of unreliable pseudolabels. At last, we can utilize some texture information such as boundaries and color based on PLP, which is always ignored in traditional approaches.
3) We take extensive experiments on the LOVEDA, Potsdam, and Vaihingen datasets to validate ACTSS. Then, some typical semisupervised semantic segmentation methods are used for comparisons. The results demonstrate that ACTSS improves accuracy effectively compared to other approaches when only a few labeled samples are available.

II. Related Work

In this section, we review the semantic segmentation in remote sensing at first. Then, we review the application of transformer in semantic segmentation. At last, we introduce the representative SSL methods.

A. Semantic Segmentation in Remote Sensing

Semantic segmentation in remote sensing is a critical task and has attracted a significant amount of research interest in recent years. Various deep learning methods have been proposed to solve this problem. One of the most popular approaches for semantic segmentation in remote sensing is CNN. For example, Mohammadimanesh et al. [25] proposed a CNN-based approach for segmentation of SAR that achieved a high accuracy. Li et al. [26] proposed a lightweight CNN with attention, called ABCNet. It can reduce the computational costs obviously. And GAN-based network was used in SSGF [27] and achieves state-of-the-art performance. Based on this, Hong et al. [28] proposed a cross-model network and achieved great success. Hu et al. [29] proposed an inverse procedural facade modeling for semantic segmentation of remote sensing image. Li et al. [30] proposed a semantic segmentation network based on weakly supervised constraints. Transformer is other method, which can be used to capture the global feature of the image. Many researchers also introduced transformer into semantic segmentation. For instance, Kwak and Kim [54] introduced a separable channel attention network (SCANet); Zhang et al. [62] introduced an additional boundary detection task to enhance the accuracy of boundary pixel positioning and reduced salt-and-pepper noise in semantic segmentation. Zhang et al. [55] developed a CNN-based multiscale feature extraction module (LFE) for small target segmentation, and Cheng et al. [51] achieved significant success with transformer-based refinement learning (TRL).

B. Transformer Network

Transformer was first proposed by Vaswani et al. [15] to deal with machine translation tasks in natural language processing (NLP). Then, transformer was applied to computer vision, such as vision transformer [31]. It has achieved outstanding performance on image classification task. Different from CNN, it can extract features hierarchically to capture long-range dependences via self-attention. By encoding the image into a sequence of patches and leveraging self-attention, the transformer model can effectively learn representations for image semantic segmentation tasks. Then, more and more scholars try to introduce transformer into vision domain. Zheng et al. [32] proposed a model based on transformer and CNN. The Swin transformer [33] presents a hierarchical structure, which further enhances its potential for semantic segmentation and achieves state-of-the-art results. Chen et al. [16] proposed a model called Transunet, which captures global information via transformer and combines it with UNet [34]. Moreover, Liu et al. [65] designed a unique module (ITM) to deal with the limitation of input size for Swin transformer and take both global and local representations into consideration.

Then, scholars started incorporating the transformer into remote sensing tasks. For instance, Xiao et al. [56] introduced a dual-branch network called EMRT, which combines CNN and transformer to extract enhanced multiscale contextual

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.

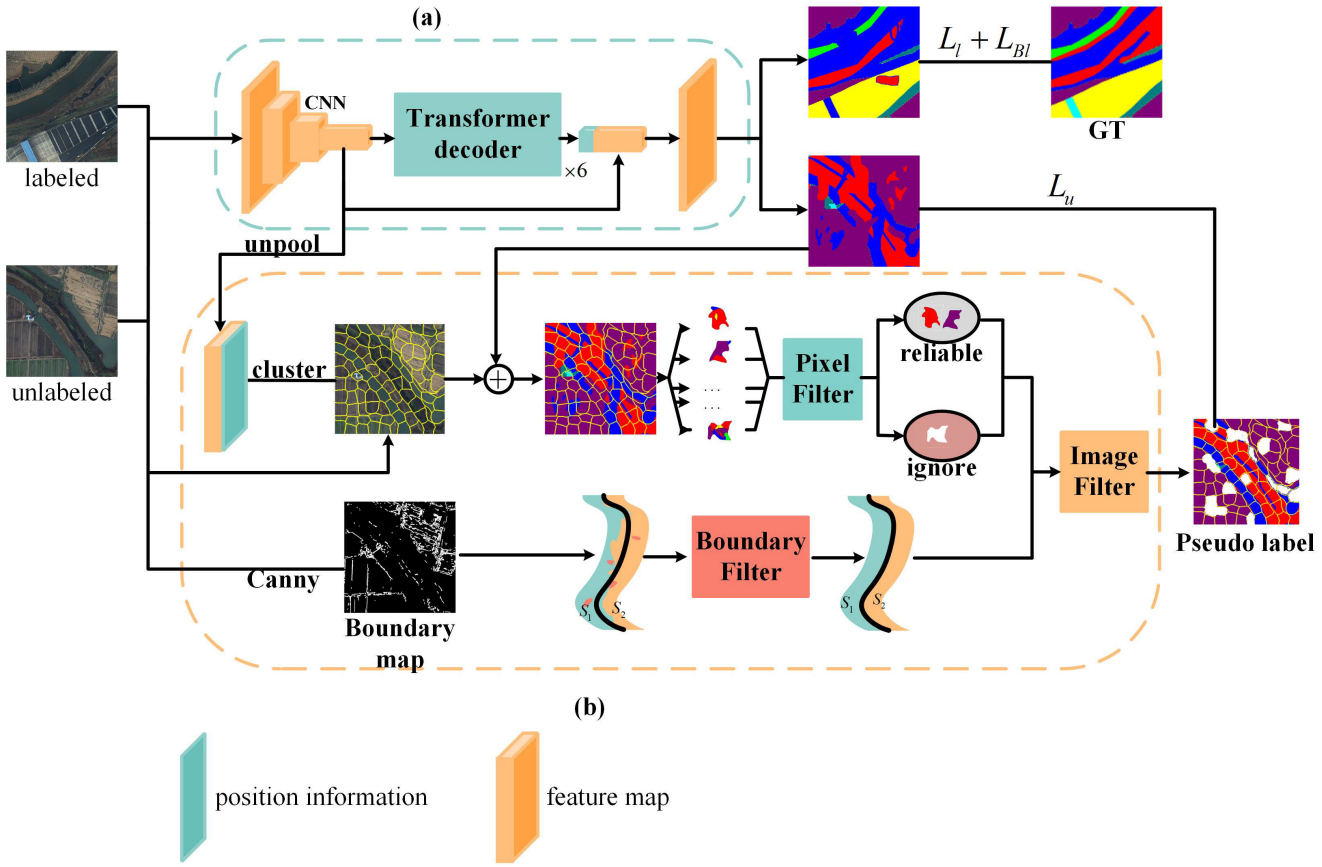


Fig. 1. Overview of the proposed ACTSS. Illustration of (a) ACT block and (b) PLP filter block. (a) is used to extract the global–local information and to segment the image. (b) is used to filter the unreliable pseudolabels to prevent them from affecting the model.

information in remote sensing. Chen et al. [57] proposed a multiresidual channel interaction attention module that utilizes attention mechanisms to learn multiscale context information. Wang et al. [58] presented a novel two-stage model ensembling framework based on the performance analysis of existing transformer-based segmentation models. Liu et al. [64] proposed feature-selective anchor mechanism (FSAM) for generating multiscale object information, enabling the extraction of aggregated features for the encoder. All of those methods have achieved great success.

C. Semisupervised Learning

Pixel-level annotation for remote sensing image is a time-costing process [35]. There is usually a lack of labels in semantic segmentation. Utilizing unlabeled data effectively through SSL can alleviate this issue [36]. For example, self-training is a basic semisupervised method. First, the model is trained on limited labeled samples and then generated pseudolabel of the unlabeled data and retrained the model on full data. Consistency regularization [37] is other important concept in SSL [42], [43]. It means that when we put small perturbations into the inputs, the output of the model should not change significantly. For example, CCT [45] put different perturbations into the input such as MixMatch [38], ReMixMatch [39], Cutout [40], and Mixup [49] and then compute the consistency loss of the different decoders. GAN-based method is other popular approach [41]. The pseudolabel predicted by model is not reliable enough. We need to select the reliable pseudolabel by GAN-based methods [44], [50]. A discriminator network was used in AdvSemiSeg [41] and S4GAN [24] to select reliable pseudolabels. To enhance semisupervised models and improve pseudolabeling in remote sensing image segmentation, Wang et al. [52] proposed ICNet, an iterative contrastive network. Huang et al. [59] introduced BSCA, a bidirectional sample-class alignment method for RS-SSDA, encompassing supervised alignment (SA) and unsupervised alignment (UA) modules. Feng et al. [60] presented an innovative approach for sample label revalidation using spatial neighborhood information (SNI). Kwak and Kim [54] utilized CycleGAN and EfficientNet to develop a semisupervised VHR remote sensing image classification method.

III. Methodology

In this section, we first propose the ACT semantic segmentation model (ACTSS). It includes two parts: a semantic segmentation network based on the ACT model and a pseudolabel filter mechanism based on clustering and boundary extraction. Then, we describe each submodule. Finally, we describe the objective function of each part.

A. ACTSS Overview

There is limited labeled data in SSL, and network with many parameters and complex structures such as CNNs can easily

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.

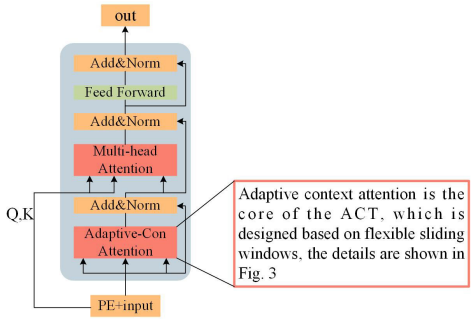


Fig. 2. Framework of the ACT.

lead to overfitting. In contrast, transformer has fewer network parameters. In this article, we introduce it into SSL to alleviate overfitting. We propose an ACT to combine global features and local features together. Moreover, to prevent incorrect pseudolabels from affecting the model, we propose a PLP filter to select reliable pseudolabels.

Fig. 1 shows the framework of our model (ACTSS). There are two major parts of ACTSS. The first one is a semantic segmentation network based on the ACT. In this module, an encoder is proposed using convolutional network to obtain features of the images. Then, in the transformer decoder, global feature and local feature are connected by adaptive context attention to enhance the generalization ability of the network. At last, we connect the outputs of CNN and transformer together by residual network and predict the segmentation.

The second part is a pseudolabel filter (PLP) mechanism based on clustering and boundary extraction. In this module, the feature map is clustered to hold similar pixels together. Then, the boundaries of the image are obtained by the Canny operator. Based on this, pseudolabels with large contradictions are filtered out in the three dimensions of point, line, and plane.

There are two branches for labeled and unlabeled samples separately. For labeled samples, we compute loss function with labels to adjust the parameters directly. For unlabeled samples, the best network in the iterative process is used to obtain pseudolabels. Then, we filter out the unreliable pseudolabels by the PLP filter. Based on those pseudolabels, we compute the loss to adjust the network.

B. Adaptive Context Transformer

The framework of the proposed model is also composed of two parts: a CNN-based encoder and a decoder based on the ACT. Fig. 2 shows the structure of the ACT. ACT can capture more small-scale features in patch token compared with the traditional shifted window in the Swin transformer, which always ignores detailed information such as colors and boundaries. Moreover, the fixed window size in the Swin transformer always limits the long-range dependences of the model. In contrast, ACT focuses more on the pixel-level feature correlation in the spatial dimension, which helps our network to boost integration further, so we propose the ACT instead of the shifted window.

ResNet is a powerful visual representation extractor, which achieves outstanding performances on ImageNet classification. ResNet-50 is set as the encoder of the model. It can obtain more contextual features of the image. Remote sensing image $x \in R^{3\times H\times W}$ is fed into the ResNet-50. At the final layer, the contextual feature $f \in R^{C_0\times H\times W}$ is produced. And then a 1×1 convolutional filter is used to reduce the channel dimension of $f_{3-\mathrm{D}} \in R^{d\times W\times H}$ from d dimension to $f_{3-\mathrm{D}}$.

The transformer architecture includes both an encoder and a decoder. However, our model skips the encoder since an encoder CNN has been used. Instead, the overall decoder consists of six transformer decoder layers; each transformer decoder layer consists of a single layer with an adaptive context attention block, a multihead attention block, and a feed-forward network (FFN).

We collect data by sliding windows with variable sizes when acquiring Q, K, and V during adaptive context attention. On the one hand, we chose a small window with the size of 16 to capture Q to extract local features. On the other hand, a bigger window with the size of 64 is used to capture K and V for global features. Then, we obtain the features of different ranges for K and V by MP. At last, we combine them together by the attention formula.

1) Adaptive Context Attention: A self-attention structure in transformers is unable to adaptively integrate global and local contexts, because the information collected by Q, K, and V sliding windows come from areas of the same size. Therefore, some long-range correlations of nonadjacent pixels or detailed information can be lost. So we introduce an adaptive context attention module to avoid the issue and consider deeper internal relationships.

Fig. 3 shows the framework of the adaptive context attention. First, we cut Q from the feature map by a sliding window with the size of 16, convert it into a 1-D vector, and code it into Q with a length of dimension. For K and V, we cut out sequences by sliding windows with size of 64 and then connect them together and obtain a 1-D vector by MP. Finally, we code them into K and V with length of dimension. Now, there are local details (such as textures, colors, and contours) in Q and global features (long-range correlations) in K/V. First, we compute the similarity between local contexts (Q) and global features (K) and then use a softmax layer to produce the global–local attention weights W by formula (1). Then, appropriately assigning W to global features (V) can fuse global and local contexts and long-range relations in a gapped manner. More attention will be allocated to the relations of global and local that have a significant impact on the result. And the final prediction result is obtained by splicing and mapping the local feature and the global attention feature to the output space.

We compute the attention by this formula after obtaining Q, K, and V

$$\mathrm{Attention}(Q, K, V) = WV = \mathrm{Soft\,max}\left(\frac{QK^T}{\sqrt{d}} + B\right)V \quad (1)$$

where Q denotes the query and B represents the relative positional bias. K and V represent the key and value of the attention and d represents the depth of the hidden layer.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.

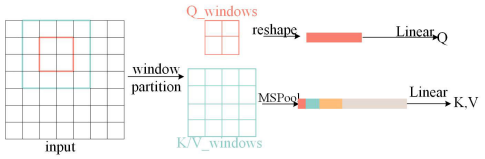


Fig. 3. Framework of the adaptive context attention.

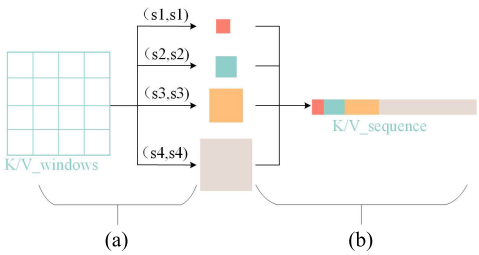


Fig. 4. Framework of MP. (a) Average pooling. (b) Flatten and concat.

2) Multiscale Pooling: Global–local features are essential in semantic segmentation. Fig. 4(a) shows the common MP methods. The sizes of the four pooled layer cores are set as 1, 3, 6, 8 and build multiple pooling layers to obtain information in different ranges using cores of different sizes (s1, s2, s3, s4). Then, as shown in Fig. 4(b), we connect them and obtain an overall K and V sequence after pooling. The lengths of the K and V sequence are the same as the Q sequence.

C. PLP Pseudofilter

Remote sensing images have lower resolution. The corresponding relationship between the pixel point and the ground object is stronger (the area corresponding to a small number of pixel points can cover the complete ground object), which is conducive to clustering at the scale of ground objects. In the proposed model, we extract the feature matrix [b, 32, 64, 64] by ResNet-50 and transfer it into feature map [b, 32, H, W] by unpooling. Then, we introduce position information (x, y) into it and get feature map [b, 34, H, W] for clustering, which is an improvement on the basis of SLIC (super-pixel segmentation). The feature map is divided into many different clusters, and we map these clusters into the original image and get the output. So the pixels with similar color, brightness, and texture are clustered into the same cluster. Theoretically, the pixels in the same cluster are most likely of the same type, and the boundary between different clusters often represents the boundary of different categories.

In addition, we extract the edge of the image by the Canny operator and regard the boundary map as another basis for pseudolabel filtering. The pseudolabel filter (PLP) mechanism based on the above characteristics is divided into three parts: point filter, boundary filter, and image filter.

1) Point Filter: Fig. 5 and formula (2) show the process of point filtering, which can be divided into two parts: filtering and absorption. The pixel points in one cluster usually belong to the same kinds of ground objects. If there are multiple categories in the corresponding cluster block, none of them has an absolute advantage in quantity. This means that the cluster is in conflict with the prediction of the network, and

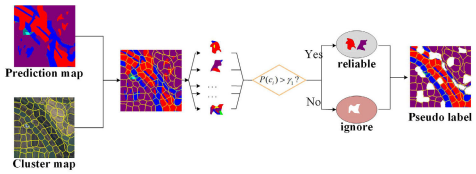


Fig. 5. Framework of the point filter.

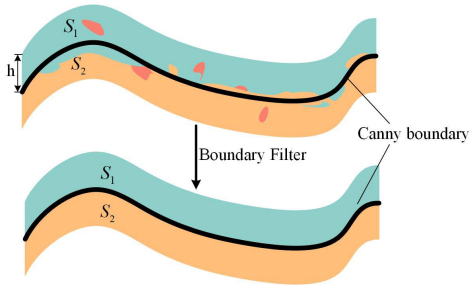


Fig. 6. Framework of the boundary filter.

the result of this cluster has a high probability of error, so it needs to be ignored. In contrast, if there is a category c_i in the cluster that has an absolute quantity advantage (for example, the frequency of the category exceeds 95%), then we can regard all points in the cluster as belonging to c_i, and the rest may be salt-and-pepper noise. The noise is absorbed by the mainstream category c_i, and the precision of the pseudolabels can be enhanced according to the point filter

$$\hat{y} = \begin{cases} c_i, & \text{if } p(c_i) > \gamma_1 \\ \text{ignore}, & \text{otherwise} \end{cases} \tag{2}$$

where γ_1 denotes the threshold of the point filter, c_i is the most frequent category in the cluster, and the frequency of c_i is $p(c_i)$.

2) Boundary Filter: Fig. 6 and formula (3) show the process of boundary filtering; the Canny operator is used to extract the boundary of the image according to the gradient change; and then, two strips S_1 and S_2 with a length of h are selected on both sides of the detected boundaries. The black line represents the boundary detected by the Canny operator. Taking strip S_1 as an example, if no other boundaries are detected in S_1, it indicates that there is no obvious gradient change in S_1. The pixel points in S_1 usually belong to the same kind of ground objects. If there is a category c_i in S_1 that has an absolute quantity advantage (for example, the frequency of c_i exceeds 95%), then we can regard all points in S_1 belonging to c_i, and the rest may be inaccurate boundaries, which will be absorbed by the mainstream category c_i. Otherwise, it will remain unchanged

$$y'_{bl} = \begin{cases} c_i, & \text{if } p(c_i) > \gamma_2 \\ \text{ignore}, & \text{otherwise} \end{cases} \tag{3}$$

where γ_2 denotes the threshold of the boundary filter. c_i is the most frequent category in the strip, and the frequency of c_i is $p(c_i)$. y'_{bl} denotes the strip on one side of the boundary (S_1 or S_2).

3) Image Filter: The image filter is composed of a point filter and boundary filter, whose purpose is to filter out unreliable

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.

prediction maps. Formula (4) shows its process. On the one hand, p_1 represents the frequency of unreliable points in point filter. When it is higher than threshold γ_1, it indicates that there is a great contradiction between the prediction map and the clustering map, so the pseudolabel is inaccurate and we should ignore it. On the other hand, p_2 represents the frequency of unreliable points in boundary filter. If p_2 is bigger than γ_2, it also means that there is a great contradiction between the prediction map and the boundary map, so the prediction map is unreliable and we should ignore it

$$y' = \begin{cases} \hat{y}, & \text{otherwise} \\ \text{ignore}, & \text{if } p_1 > \gamma_1 || p_2 > \gamma_2 \end{cases} \tag{4}$$

where γ_1 denotes the threshold of the point filter. γ_2 denotes the threshold of the boundary filter.

D. Objective Function

The image dataset of remote sensing provided in the SSL includes unlabeled data and labeled data. Let $U = \{u_i\}$ denote the M unlabeled samples. u_i is the ith unlabeled input image. N labeled samples are represented as $X = \{x_i, y_i\}$. x_i is the image and y_i is the corresponding label. The main task of SSL is to train a segmentation model based on these data. There are two loss functions in this article: the first one is cross-entropy loss function, whose formula is as follows:

$$H(y_i, f(x_i, \theta)) = -\frac{1}{W * H} \sum_{i=1}^{w} \sum_{j=1}^{h} \sum_{c=1}^{C} y_{i,j,c} \log f(x, \theta) \tag{5}$$

where $H(y_i, f(x_i, \theta))$ denotes the loss of each pixel, while H and W represent the input image's height and width.

Formula (6) is the boundary loss function, which is used for semantic image segmentation that calculates the difference between predicted and ground-truth boundaries. The formula is as follows:

$$B_i = \begin{cases} 1, & \text{if } \exists \ \mathrm{KL}(P_i, P_j) > \varepsilon, \ \ j \in N_2(i) \\ 0, & \text{Otherwise} \end{cases} \tag{6}$$

where P_i is the C-dimensional vector extracted from the probability map at pixel i. $N_2(i)$ indicates the two-neighborhood of pixel i. For labeled data $X = \{x_i, y_i\}$, $f(x, \theta_1)$ is used to obtain the prediction and then compute the loss

$$L_l = \frac{1}{X} \sum_{x \in X} H(y_x, f(x, \theta_1)). \tag{7}$$

Moreover, we calculate the boundary loss with the label y_x

$$L_{Bl} = \frac{1}{X} \sum_{x \in X} H_b(y^b{}_x, y_x). \tag{8}$$

For unlabeled data $U = \{u_i\}$, $f(x, \theta)$ is used to generate the prediction map

$$y_u = f(u_i, \theta_1). \tag{9}$$

Then, the PLP filter $g(\theta_2)$ is used to filter out unreliable pixels and boundaries. At the same time, regions and boundaries with high reliability are optimized

$$y' = g(y_u, \theta_2). \tag{10}$$

Algorithm 1 Adaptive Context Transformer for Semisupervised (ACTSS) Segmentation

Input: labeled data X, unlabeled data U, parameters $\lambda_a, \lambda_b, \gamma_1, \gamma_2$;
Initialize: $\theta, \omega, t = 1, T = 60, mIoU_{best} = 0, \sigma = 1 \times 10^{-3}$;
Procedure:
While $t \leq T$:
    predict segmentation results $f(x, \theta)$ of the labeled image;
    calculate the mIoU of the model;
    if mIoU $> mIoU_{best}$ then;
        $mIoU_{best}$ = mIoU;
        Generate pseudolabel of unlabeled data by $y_u = \arg\max_c(f(u_x, \theta_b))$;
        Filtering y_u by pseudolabel Filter (PLP);
    else:
        pass;
    end if;
    calculate loss L of unlabeled and labeled samples;
    update θ of the model;
Output: the parameters θ of the model
Update: t = t + 1
End While

Then, we obtain the pseudolabel $P = \{x_u, y'_u\}$; here is the formula for the loss function when working with unlabeled data

$$L_u = \frac{1}{U} \sum_{x \in U} H\left(y'_u, f(x_u, \theta_1)\right). \tag{11}$$

Here is the formula for the overall loss function when working with both labeled and unlabeled data

$$L = L_l + \lambda_a * L_u + \lambda_b * L_{Bl}. \tag{12}$$

The weight of the unlabeled loss is indicated by λ_a, and the weight of the boundary loss is indicated by λ_b. The parameter θ of the network $f(x, \theta)$ is adjusted according to L. The training process of ACTSS is shown in Algorithm 1.

IV. Experiments

In this section, we evaluate the performance of the proposed ACTSS approach on three public datasets: the LOVEDA, Potsdam, and Vaihingen datasets. Then, we compare it with the related semisupervised semantic segmentation methods.

A. Implementation Setting

1) Dataset:

a) LOVEDA: LOVEDA dataset is an adaptive surface coverage dataset in China's urban and rural areas, which consists of 5987 remote sensing images with high spatial resolution (0.3 m), including urban and rural landscapes in Nanjing, Changzhou, and Wuhan. The images are 1024 × 1024 pixels with annotations for seven semantic classes: agriculture, background, forest, barren, water, road, and buildings.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.

TABLE I
NUMBER OF LABELS WITH DIFFERENT RATIOS

Datasets	Labeled Data Amount				
	1/16	1/8	1/4	1/2	Full
Potsdam	117	235	470	940	1880
Vaihingen	115	230	460	920	1840
LOVEDA	180	360	720	1440	2880

b) Potsdam: Potsdam dataset contains 38 remote sensing images collected from the typical historic city Potsdam. It includes six common land-cover classes: cars, backgrounds, trees, low vegetation, buildings, and impervious surfaces. Each image is with a size of 6000 × 6000 pixels. Limited by the GPU memory, we cut it into 512 × 512 pixels for training.

c) Vaihingen: Vaihingen dataset is a collection of 33 aerial images with a spatial resolution of 9 cm. The images were taken over the city of Vaihingen and have an average size of over 2000 × 2000 pixels. The dataset consists of three bands, namely near infrared, red, and green bands, and it includes six categories: cars, backgrounds, trees, low vegetation, building, and impervious surfaces. We cut it into 512 × 512 pixels for training.

In the experiment, we select 1/2, 1/4, 1/8, and 1/16 of the entire dataset as labeled data. Then, we left the remaining data unlabeled. The image number of different ratios is shown in Table I. For all dataset, unused data are remained for validation and testing, accounting for 15% and 20% of the total data, respectively.

2) Implementation Detail: All experiments are implemented on Python 3.10.1. When using the LOVEDA, Potsdam, and Vaihingen datasets to train the model, the learning rate is initially set as 10^{-3}, and the training epoch is set as 60. The learning rate is multiplied by $(1-(\text{iter}/\max \text{iter}))^{0.9}$ in every iteration. The SGD with momentum is adapted. The momentum is fixed at 0.9, and the weight decay is 10^{-4}. Due to the limitation of the GPU, the batch size for LOVEDA is set as 2 and that for Potsdam and Vaihingen is set as 4.

3) Evaluation Metrics: The evaluation metrics used in our experiments included two major categories. The first is the mean intersection over union (mIoU), the average value of IoU, which can be formulated as

$$\text{IoU} = \frac{\left|R_g \cap R_p\right|}{\left|R_g \cup R_p\right|} = \frac{\left|R_g \cap R_p\right|}{\left|R_g\right| + \left|R_p\right| - \left|R_g \cap R_p\right|} \tag{13}$$

where R_g and R_p represent label's pixel points and prediction's pixel points, respectively. $\cap$ and $\cup$ represent intersection and union, respectively. $|\cdot|$ indicates the total number of pixels in this collection. The other one is precision accuracy (PA), which can be formulated as

$$\text{PA} = \frac{P_r}{P_b} \tag{14}$$

where P_b represents the total number of labeled pixels in the label and P_r represents the number of pixels that are the same as the label in the prediction map.

4) Models for Comparison: We compare our model with several common semantic segmentation methods. Examples include guided collaborative training (GCT) [45], unified dual-stream perturbations approach (UniMatch) [53], self-training plus plus (ST++) [47], cross pseudosupervision (CPS) [44], CCT [45], mean-teacher (MT) [48], UNet-like transformer (UNetFormer) [63], etc. For fair comparisons, all segmentation networks of those methods are based on ResNet-50.

1) *UniMatch* [53]*:* This algorithm propose a unified perturbation framework that unifies image-level and feature-level perturbations in independent streams and design a dual-stream perturbation strategy to fully probe predefined image-level perturbation space.
2) *ST++* [47]*:* This algorithm is based on ST and gradually utilizes unlabeled images from easy to difficult. In contrast, ST selects pixels with high confidence. At first, ST++ selects reliable images according to the stability of pseudolabels and then trains the model on the existing labeled samples and pseudolabels.
3) *CPS* [44]*:* It sends the same image to two different initialization networks in this algorithm, and the pseudolabels generated from one branch are used to monitor the other branch. Back propagation instead of the moving average is used for optimization in the training process.
4) *CCT* [45]*:* Consistency constraints are used in this algorithm to solve the semisupervised problem, which trains the encoder and main decoder of shared parameters on unlabeled samples and labeled samples, respectively. It adds two auxiliary detection branches Aux Decoder1 and Aux Decoder2 to the unlabeled data training model. Then, consistency constraints are applied to the main decoder, Aux Decoder1 and Aux Decoder2.
5) *GCT* [46]*:* This algorithm solves the problem caused by dense output through a novel defect detector and learns from unlabeled data through two newly proposed constraints independent of specific task attributes.
6) *MT* [48]*:* It divides the model into teachers and students. The teachers generate students' learning goals when the students use the goals provided by the teachers to learn. The weight of the teacher model is obtained by the weighted average of the time memory of the student model. Moreover, the algorithm is also based on consistency regularization. It is believed that when a small disturbance noise is added to the input data, the prediction results will not change.
7) *UNetFormer* [63]*:* It proposes a transformer-based decoder, constructs a UNetFormer for real-time urban scene segmentation, and develops an efficient global–local attention mechanism to model both global and local information in the decoder.

B. Comparison With the Related Methods

We compare the proposed model with some semisupervised semantic segmentation methods in this section. Implement these methods on the Potsdam, Vaihingen, and LOVEDA datasets and then analyze the advantages of ACTSS over these methods.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON RESULTS ON THE POTSDAM DATASET

Methods	Ratio of labeled samples							
	1/16		1/8		1/4		1/2	
	mIoU	Precision	mIoU	Precision	mIoU	Precision	mIoU	Precision
Sup	76.54	86.48	79.65	89.48	82.74	92.13	84.23	93.37
UNetFormer	77.92	**90.22**	82.24	90.31	84.47	94.17	85.03	94.85
GCT	77.93	87.46	78.84	90.81	82.48	93.12	83.83	93.72
MT	77.14	88.83	78.55	90.55	82.73	93.97	84.27	94.54
CCT	77.72	88.44	78.38	89.32	83.27	92.08	83.84	93.58
CPS	78.25	88.45	80.42	88.32	82.52	93.42	84.84	94.51
ST++	77.44	87.23	79.55	90.95	83.56	93.33	84.83	93.72
UniMatch	78.44	89.13	81.14	**91.94**	84.03	93.65	85.01	94.05
Ours	**79.63**	89.3	**82.47**	90.42	**84.17**	**94.51**	**85.34**	**95.38**

TABLE III
COMPARISON RESULTS ON THE VAIHINGEN DATASET

Methods	Ratio of labeled samples							
	1/16		1/8		1/4		1/2	
	mIoU	Precision	mIoU	Precision	mIoU	Precision	mIoU	Precision
Sup	74.73	82.24	77.92	85.62	81.47	88.26	83.54	92.91
UNetFormer	76.28	87.84	79.29	**90.32**	82.69	91.15	85.06	93.41
GCT	75.33	83.15	77.93	87.61	78.24	89.55	83.93	94.83
MT	75.74	83.76	76.93	87.82	79.57	90.15	82.87	93.16
CCT	75.76	84.15	78.86	86.43	82.33	91.83	83.02	94.64
CPS	76.03	85.63	79.34	87.64	82.02	90.42	84.19	93.37
ST++	75.31	85.68	78.28	86.46	80.04	91.75	83.95	94.73
UniMatch	76.71	86.47	79.44	89.93	81.15	91.03	84.72	95.15
Ours	**77.98**	**87.23**	**80.38**	88.17	**82.81**	**92.34**	**84.83**	**95.45**

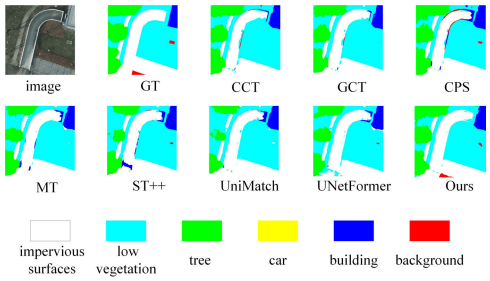


Fig. 7. Representative competitive results on the Potsdam dataset at a labeled ratio of 1/4.

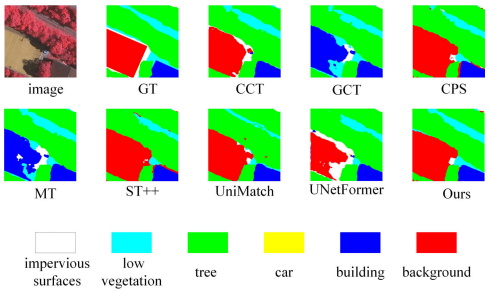


Fig. 8. Representative competitive results on the Vaihingen dataset at a labeled ratio of 1/4.

1) Comparisons on Potsdam: As shown in Table II, we compare the quantitative results on the Potsdam dataset. The supervised-only method, in which only labeled samples are used, shows the poorest performance and suffers even more when the labeled samples decrease. ACTSS, on the other hand, achieves similar or better results compared to other methods. For instance, it outperforms the CPS by over 1.37% and 0.85% regarding the mIoU and precision when processing the labeled amount 1/16, which is much higher than other semisupervised frameworks. Fig. 7 displays some representative results from all competitive methods. From Fig. 7, it is evident that CCT is not suitable for the classification of slender objects; GCT and CPS often struggle to differentiate between background and impervious surfaces; MT is not adept at extracting small ground objects; ST++ and UNetFormer have difficulty in extracting the boundaries of ground objects and are prone to generating noise in large buildings. The effect of UniMatch in identifying the edges of ground objects is slightly poor. ACTSS, however, has good continuity in segmenting long and thin objects and is ideal for boundary extraction. Additionally, the PLP filter is effective in reducing salt-and-pepper noise of the prediction map.

2) Comparisons on Vaihingen: Then, the comparisons between the ACTSS and others on the Vaihingen dataset is reported in Table III. Both mIoU and precision are compared in this table. Our approach demonstrates better performance when fewer labeled samples are used. The advantage decreases as the labeled samples increases. For instance, our approach outperforms CPS by 1.95% and 1.03% in mIoU under 1/16 and 1/8 partition protocols. The increases under high-partition protocols 1/4 and 1/2 are only minor, at 0.79% and 0.64%, respectively. Fig. 8 shows partial visual comparison results of all the methods. CCT has difficulty extracting small low vegetation (cyan), easily confusing low vegetation and trees. GCT and MT are prone to confusing buildings (blue) and backgrounds (red). CPS, UniMatch, and ST++ have a common drawback, which is that noise is easily generated at the boundaries of buildings. UNetFormer always

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON RESULTS ON THE LOVEDA DATASET

Methods	Ratio of labeled samples							
	1/16		1/8		1/4		1/2	
	mIoU	Precision	mIoU	Precision	mIoU	Precision	mIoU	Precision
Sup	61.16	79.43	64.87	82.52	67.33	84.87	69.72	87.51
UNetFormer	62.48	81.29	64.79	83.36	67.53	88.73	69.83	88.36
GCT	59.83	80.94	63.45	84.25	65.14	86.43	67.34	88.64
MT	60.34	80.52	62.44	84.62	65.15	86.12	68.13	87.24
CCT	61.56	80.73	63.35	83.43	66.22	86.81	69.8	88.46
CPS	59.75	81.43	63.83	85.86	66.62	88.62	67.16	88.17
ST++	60.42	80.73	63.67	85.37	65.86	87.95	69.23	89.12
UniMatch	62.17	82.45	64.55	**85.94**	67.72	88.39	69.32	89.15
Ours	**64.22**	**83.92**	**67.34**	84.4	**68.77**	**88.83**	**70.44**	**89.3**

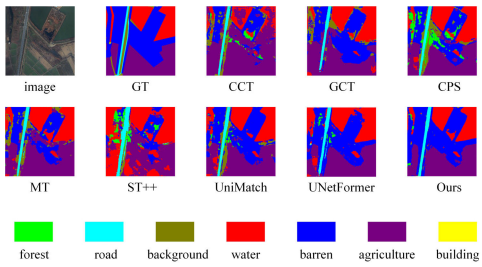


Fig. 9. Representative competitive results on the LOVEDA dataset at a labeled ratio of 1/4.

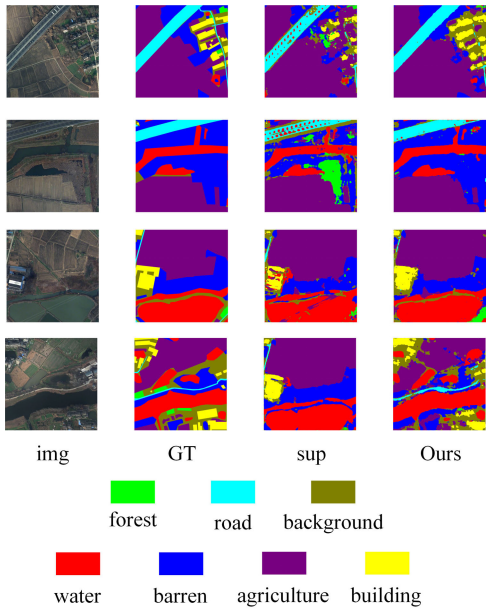


Fig. 10. Visualization results on the LOVEDA dataset (1/8 labeled samples).

confuses background and impervious surfaces. On the other hand, ACTSS has strong continuity when segmenting slender objects, can better identify boundaries, and has relatively less salt-and-pepper noise.

3) Comparisons on LOVEDA: Experimental results on the LOVEDA datasets are shown in Table IV. UNetFormer is found to be the best method among the comparison methods. Compared with UNetFormer in mIoU and precision, ACTSS slightly increases by 0.61% and 0.94% under high partition (1/2). The advantage increases when the labeled ratio decreases. It increases be approximately 1.74% and 2.62% under low partition protocols (1/16) when UNetFormer is compared. ACTSS outperforms other approaches with evaluation metrics of 64.22% and 83.92% under 1/16 partition protocols, proving its substantial advantages under low-partition protocols. Fig. 9 shows partial visual comparison results from all the methods, and it is worth noting that the proposed method produces accurate predictions compared to other semisupervised segmentation frameworks, especially in long and thin categories that are controversial or difficult to classify, such as the road, barren, and agriculture. For example, it is difficult for CCT and UniMatch to extract long and thin curved roads; GCT, CPS, and MT easily confuse the background and road; in addition, noise easily appears in large barren (blue) areas of GCT and CPS; and ST++ easily confuses agricultural land and wasteland. UNetFormer struggles to differentiate between water and barren. Of all the methods, only ACTSS has strong continuity in road recognition and can segment barren areas well.

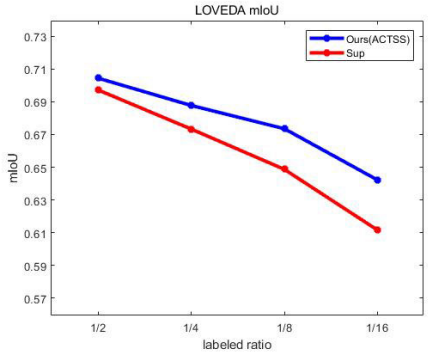


Fig. 11. Comparison with the supervised-only method on the LOVEDA dataset.

From the experiments above, our research indicates that ACTSS performs best among all partition protocols, especially in cases where labeled ratio is low. Its advantages are quite significant in such scenarios. The advantage will decrease when the proportion of labeled data increases. Preliminary analysis shows that the segment network can reach a high accuracy with more labels. If we continue to filter out pseudolabels by PLP, it is likely to filter out many correct labels, thus inhibiting the effect of the model.

C. Effectiveness Study

In order to validate the effectiveness of unlabeled data in ACTSS, we have designed comparative experiments on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.

TABLE V
MODEL'S EXPERIMENTAL RESULTS (MIOU AND IOU) ON THE LOVEDA DATASET

Partition	Methods	mIoU	building	road	water	barren	forest	farmland	others
1/2	Sup	69.72	69.82	61.81	84.56	60.63	63.26	82.52	65.41
	ACTSS	70.44	70.92	65.08	89.75	56.84	65.62	80.96	63.9
1/4	Sup	67.33	65.63	61.88	81.23	58.63	60.02	78.61	65.31
	ACTSS	68.77	65.07	64.95	88.58	57.51	62.1	78.74	64.43
1/8	Sup	64.87	65.11	57.6	80.04	54.81	60.13	75.55	60.85
	ACTSS	67.35	67.55	62.45	85.3	55.51	59.69	77.82	63.13
1/16	Sup	61.16	61.65	54.6	72.31	55.31	55.21	70.7	58.34
	ACTSS	64.22	65.51	61.61	79.63	56.82	52.84	73.88	59.25

three different datasets in this article. In one group, both labeled and unlabeled data are used, while only labeled data are used in the other group (sup). We present visualization examples of the LOVEDA validation set in Fig. 10. It shows that there is more salt-and-pepper noise when roads are segmented by the supervised-only method. The continuity is weak, and the boundaries are rough. The errors are corrected by the PLP filter of the proposed ACTSS approach, and the predictions are more accurate than those of the supervised-only approach.

We have observed significant enhancements over the supervised-only method on the LOVEDA dataset, as shown in Fig. 11. Our proposed approach consistently outperforms the supervised-only model across all partition protocols, and the gains of the ACTSS increase as the ratio of labeled data decreases. To investigate the improvements further, we provide detailed quantitative results in Table V. The supervised-only (Sup) method means that only labeled data are used for training, and ACTSS means that all data are used. Our approach shows gains of 3.06%, 2.47%, 1.44%, and 0.72% when 1/16, 1/8, 1/4, and 1/2 labeled samples are used, respectively; the utilization of the unlabeled data allows the model to achieve better performance than when only supervised learning is used. From different categories, we can know that the improvements for road and water are obvious, achieving 7.00% and 7.31% increases under the 1/16 partition protocols, respectively. Preliminary analysis may be due to the obvious boundary of the road, which is conducive to extracting the boundary by Canny. The color similarity of the same water area is conducive to clustering, and these two characteristics are conducive to the role of the PLP filter.

We evaluate ACTSS on the Potsdam dataset. Fig. 12 shows the mIoU of ACTSS and supervised-only method under all ratios of labeled data. From Fig. 12, we can know that the advantage of ACTSS is more obvious when fewer samples are used. Table VI shows the quantitative results of all kinds of categories. It shows that our model's performance improves across all metrics due to the utilization of the unlabeled samples. The supervised-only (Sup) means that only labeled data are used for training, and ACTSS means that the model is trained on all data. Our method achieves a performance improvement of 1.10% when 1/2 labeled data are used. The advantage increases when the labeled ratio decreases. The mIoU increases to 3.08% over the baseline under the 1/16 partition protocol. Notably, the tree class group shows the most substantial improvement,

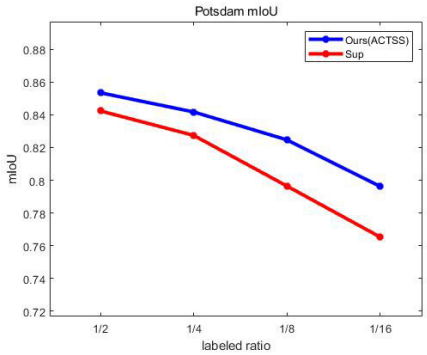


Fig. 12. Comparison with the supervised-only method on the Potsdam dataset.

reaching up to 7.70% with only 1/16 labeled data are used.

Different from the Potsdam dataset, which focuses on urban scenes, the Vaihingen dataset mainly comes from rural scenes. We evaluate ACTSS on it. Table VII and Fig. 13 show the experimental results compared with supervised-only (Sup) method. Sup means that only labeled samples are used, and ACTSS means that all data are used for training. Similar to Potsdam, ACTSS achieves more advantage compared with the supervised-only method. The advantage of performance becomes more obvious when fewer labels are utilized. Table VII shows the results when there are label ratios of 1/16, 1/8, 1/4, and 1/2. The increase in mIoU was 3.25%, 2.45%, 1.34%, and 1.28%, respectively. The improvement for building is most obvious, achieving an 8.55% increase under the 1/16 partition protocol.

D. Ablation Study

1) Parameter Analysis: The effectiveness of different parameters of the ACTSS is further evaluated on the Potsdam validation dataset when 1/4 labeled samples are used. Table VIII shows the quantitative results. There are two important thresholds for pseudofilter: threshold γ_1 of the point filter and threshold γ_2 of the boundary filter. A large number of correct pixel points will be filtered out if the threshold is set higher; otherwise, on the contrary. Too many incorrect points will be retained to mislead the training if the threshold is set lower. Through many experiments, we selected the best thresholds γ_1 and γ_2. Table VIII shows the impact of different thresholds on the model when 1/4 labeled Potsdam data are used. Through many experiments, it is evident that the model's predictive accuracy is optimal when both thresholds are set to 0.95.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
MODEL'S EXPERIMENTAL RESULTS (MIOU AND IOU) ON THE POTSDAM DATASET

Partition	Methods	mIoU	back-ground	car	tree	low vegetation	building	impervious surfaces
1/2	Sup	84.23	67.03	85.59	81.65	85.59	96.51	90.01
	ACTSS	85.34	68.83	86.51	83.92	83.5	97.45	91.83
1/4	Sup	82.74	64.96	85.23	82.99	80.21	93.47	89.58
	ACTSS	84.17	65.65	84.73	87.55	82.99	93.92	90.18
1/8	Sup	79.63	60.8	80.34	79.58	78.82	93.07	85.17
	ACTSS	82.47	60.32	82.65	87.05	82.6	94.18	88.02
1/16	Sup	76.54	56.74	80.14	71.96	75.88	88.82	85.7
	ACTSS	79.63	59.13	82.04	79.66	76.83	93.66	86.46

TABLE VII
MODEL'S EXPERIMENTAL RESULTS (MIOU AND IOU) ON THE VAIHINGEN DATASET

Partition	Methods	mIoU	back-ground	car	tree	low vegetation	building	impervious surfaces
1/2	Sup	83.54	80.14	70.33	84.47	83.29	92.78	87.23
	ACTSS	84.83	81.96	72.04	86.24	80.89	95.71	92.14
1/4	Sup	81.47	75.2	70.73	82.54	80.52	92.68	87.16
	ACTSS	82.81	79.67	71.79	82.54	79.53	94.92	88.41
1/8	Sup	77.92	74.36	66.36	78.54	74.26	88.64	83.36
	ACTSS	80.38	76.93	69.82	81.55	75.96	92.76	85.26
1/16	Sup	74.73	72.55	65.66	75.27	72.84	81.4	80.66
	ACTSS	77.98	74.93	67.55	77.66	75.66	89.95	82.13

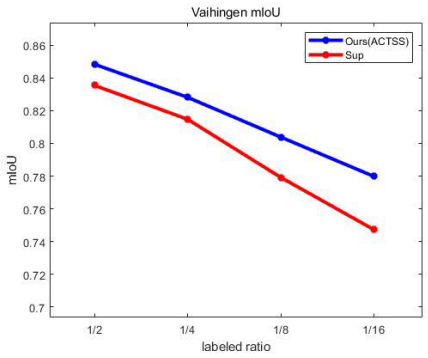


Fig. 13. Comparison with the supervised-only method on the Vaihingen dataset.

TABLE VIII
EFFECTIVENESS OF TWO FILTER THRESHOLDS ON THE SEGMENTATION RESULTS

γ_1 \ γ_2	0.98	0.95	0.9	0.8
0.98	82.37	83.66	82.15	80.82
0.95	83.19	84.17	81.32	81.43
0.9	80.33	83.17	82.17	81.85
0.8	79.68	81.56	82.56	79.27

TABLE IX
QUANTITATIVE RESULTS ON THE POTSDAM DATASET UNDER 1/4 PARTITION PROTOCOL

PLP Filter			Attention		mIoU
Point Filter	Boundard Filter	Image Filter	Adaptive Context Attention	Self Attention	
				✓	78.12
			✓		81.04
	✓		✓		82.31
✓			✓		82.54
✓	✓		✓		83.48
✓	✓	✓		✓	82.59
✓	✓	✓	✓		84.17

2) Impact of Each Submodule: To assess the performance of different modules, we have conducted additional experiment on the Potsdam dataset under the 1/4 partition protocol. Table IX shows the experimental results of PLP (point filter, boundary filter, and image filter), and we compare the performance of adaptive context attention and self-attention. We evaluate the effectiveness individually under each of them.

Both PLP filter and adaptive context attention were found effective in improving the performance of the ACTSS. As for PLP filter, by adopting the point filter, the mIoU was improved from 81.04% to 82.54%. Filtering out unreliable pixels results in an improvement of 1.50%. The pseudolabel is not reliable enough, since the Point Filter only focuses on the pixel, which limits the improvement. In contrast, the boundary filter improves boundary quality from 81.04% to 82.31%. Then, when both the point filter and boundary filter were adopted, the mIoU improves from 81.04% to 83.48%. Finally, we introduced point filter, boundary filter, and image filter into the experiment. The three filters act in three different ways: point, line, and plane. Their collaboration improves the mIoU from 81.04% to 84.17%. Addressing global–local features of the image, we compared the adaptive context attention with self-attention. Compared with self-attention, the mIoU of the adaptive context attention improved from 78.12% to 81.04% without PLP filter. And the mIoU improved from 82.59% to 84.17% when PLP filter was introduced. From the experiment, we can know that the improvement of the point filter is more obvious, because the determination of the pseudolabels for the agreement parts is pixel-level, which is more likely influenced by the noisy label. Benefitting from both the proposed point filter and the proposed ACT, the mIoU of ACTSS achieves 84.17%, overperforming the baseline by 6.04%.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.

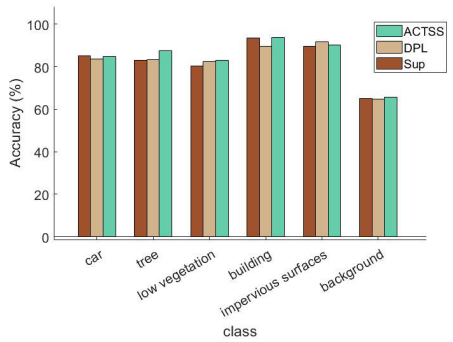


Fig. 14. Effectiveness of different categories on the Potsdam dataset.

TABLE X
COMPARISON OF THREE PSEUDOLABEL FILTERS ON THE POTSDAM DATASET UNDER THE 1/4 PARTITION PROTOCOL

Pseudo label Strategy	mIoU (%)
Pixel-wise confidence	82.04
Class-wise confidence	82.42
image-level confidence	81.36
direct use of unrefined pseudo-labels	80.07
Ours (PLP Filter)	84.17

The proposed method is further compared with the mutual pseudosupervision on the Potsdam dataset under the 1/4 partition protocol, which directly utilizes the predictions to generate the pseudolabels and applies pseudosupervision based on that. Fig. 14 shows the performances on the six class groups. From Fig. 14, we can know that the performance of the ACTSS is better than direct mutual pseudosupervision on most categories. The improvement for trees is most obvious, which are similar in color, so it is easy to cluster. PLP can play a better role for trees. We can get more reliable pseudolabels benefitting from PLP filter. It results in obvious improvements on all categories.

Finally, we analyze the quantitative results of different pseudofilters. Four typical methods are compared, including the pixelwise confidence, classwise confusion, image-level confidence, and direct use of unrefined pseudolabels. Table X shows the results. The pixelwise confidence indicates that a variable threshold is set to judge the reliability of the points in the prediction. A pseudofilter with a higher threshold is eager to regard the correct pixels as wrong pixels; in contrast, a lower threshold is eager to regard the wrong pixels as correct results, and both are harmful to the result. Classwise confidence represents a dynamic threshold that is set for each type. In classwise confidence, we give different thresholds to each class according to the difficulty of the category, and these thresholds can be adjusted according the training. Image-level confidence denotes the confidence level of the pseudolabels. The mIoU of the model achieves 82.04% or 82.42% when pixelwise confidence and classwise confusion are used, so when the PLP filter is used on the proposed model, the mIoU increases by 2.21% and 1.73%. Compared with image-level confidence and direct use of unrefined pseudolabels, the mIoU of the PLP filter increases by 2.81% and 4.1%, respectively, which shows the advantage of the PLP filter.

V. CONCLUSION AND FUTURE WORK

In this article, we propose an adaptive context attention semisupervised semantic segmentation (ACTSS). The model is composed of two parts. The first one is transformer semantic segmentation network based on adaptive context attention, which can extract global–local features based on flexible sliding windows for K and V. The second one is point–line–plane (PLP) filter mechanism based on clustering and boundary extraction, which can select reliable pseudolabels from the predict map.

We have evaluated the proposed approach on three widely used datasets: the LOVEDA, Potsdam, and Vaihingen datasets. Extensive experimental results demonstrate the effectiveness of the ACTSS and demonstrate its state-of-the-art performance for semisupervised semantic segmentation, especially when labels are limited. Preliminary analysis may be due to the high-reliable label reducing the advantage of the PLP filter when more labels are used. Therefore, the evaluation of the PLP filter on more labeled images deserves further study.

REFERENCES

[1] Y. Zou et al., "PseudoSeg: Designing pseudo labels for semantic segmentation," 2020, *arXiv:2010.09713*.
[2] Q. Zhao, J. Liu, Y. Li, and H. Zhang, "Semantic segmentation with attention mechanism for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5403913.
[3] Q. Zhang, G. Yang, and G. Zhang, "Collaborative network for super-resolution and semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4404512.
[4] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
[5] G. French, S. Laine, T. Aila, M. Mackiewicz, and G. Finlayson, "Semi-supervised semantic segmentation needs strong, varied perturbations," 2019, *arXiv:1906.01916*.
[6] J. Al-Doski, F. M. Hassan, M. Norman, and A. A. Najim, "Interaction of image fusion techniques and atmospheric correction for improve SVM accuracy," *Earth Sci. Informat.*, vol. 15, no. 4, pp. 2673–2687, Dec. 2022.
[7] J. Feng, D. Wang, and Z. Gu, "Bidirectional flow decision tree for reliable remote sensing image scene classification," *Remote Sens.*, vol. 14, no. 16, p. 3943, Aug. 2022.
[8] A. K. Heidinger, A. T. Evan, M. J. Foster, and A. Walther, "A naive Bayesian cloud-detection scheme derived from CALIPSO and applied within PATMOS-x," *J. Appl. Meteorol. Climatol.*, vol. 51, no. 6, pp. 1129–1144, Jun. 2012.
[9] S. M. Jaisakthi, P. Mirunalini, and C. Aravindan, "Automated skin lesion segmentation of dermoscopic images using GrabCut and k-means algorithms," *IET Comput. Vis.*, vol. 12, no. 8, pp. 1088–1095, Dec. 2018.
[10] H. Lu, H. Su, P. Zheng, Y. Gao, and Q. Du, "Weighted residual dynamic ensemble learning for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6912–6927, 2022.
[11] K. Ronald, R. Luu, and C. Kanan, "Low-shot learning for the semantic segmentation of remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6214–6223, Oct. 2018.
[12] N. Wambugu et al., "Hyperspectral image classification on insufficient-sample and feature learning using deep neural networks: A review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 105, Dec. 2021, Art. no. 102603.
[13] P. Zhang, B. Zhang, T. Zhang, D. Chen, and F. Wen, "Robust mutual learning for semi-supervised semantic segmentation," 2021, *arXiv:2106.00609*.
[14] Y. Zou, Z. Yu, B. V. K. V. Kumar, and J. Wang, "Unsupervised domain adaptation for semantic segmentation via class-balanced self-training," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 289–305.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.

[15] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 6000–6010.

[16] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.

[17] M. S. Ibrahim, A. Vahdat, M. Ranjbar, and W. G. Macready, "Semi-supervised semantic image segmentation with self-correcting networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12715–12725.

[18] R. Mendel, L. A. de Souza Jr., D. Rauber, J. P. Papa, and C. Palm, "Semi-supervised segmentation based on error-correcting supervision," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K. Berlin, Germany: Springer, Aug. 2020, pp. 141–157.

[19] C. Li, T. Xu, J. Zhu, and B. Zhang, "Triple generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 4091–4101.

[20] Y. Luo, J. Zhu, M. Li, Y. Ren, and B. Zhang, "Smooth neighbors on teacher graphs for semi-supervised learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8896–8905.

[21] J. T. Springenberg, "Unsupervised and semi-supervised learning with categorical generative adversarial networks," 2015, *arXiv:1511.06390*.

[22] J. Zhang, Z. Li, C. Zhang, and H. Ma, "Robust adversarial learning for semi-supervised semantic segmentation," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 728–732.

[23] S. Mittal, M. Tatarchenko, and T. Brox, "Semi-supervised semantic segmentation with high- and low-level consistency," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1369–1379, Apr. 2021.

[24] S. Fan, F. Zhu, Z. Feng, Y. Lv, M. Song, and F.-Y. Wang, "Conservative-progressive collaborative learning for semi-supervised semantic segmentation," *IEEE Trans. Image Process.*, early access, Feb. 10, 2023, doi: 10.1109/TIP.2023.3242819.

[25] F. Mohammadimanesh, B. Salehi, M. Mahdianpari, E. Gill, and M. Molinier, "A new fully convolutional neural network for semantic segmentation of polarimetric SAR imagery in complex land cover ecosystem," *ISPRS J. Photogramm. Remote Sens.*, vol. 151, pp. 223–236, May 2019.

[26] R. Li, S. Zheng, C. Zhang, C. Duan, L. Wang, and P. M. Atkinson, "ABCNet: Attentive bilateral contextual network for efficient semantic segmentation of fine-resolution remotely sensed imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 181, pp. 84–98, Nov. 2021.

[27] W. Han, R. Feng, L. Wang, and Y. Cheng, "A semi-supervised generative framework with deep learning features for high-resolution remote sensing image scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 23–43, Nov. 2018.

[28] D. Hong, N. Yokoya, G.-S. Xia, J. Chanussot, and X. X. Zhu, "X-ModalNet: A semi-supervised deep cross-modal network for classification of remote sensing data," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 12–23, Sep. 2020.

[29] H. Hu et al., "Semi-supervised adversarial recognition of refined window structures for inverse procedural façade modelling," *ISPRS J. Photogramm. Remote Sens.*, vol. 192, pp. 215–231, Oct. 2022.

[30] Y. Li, T. Shi, Y. Zhang, W. Chen, Z. Wang, and H. Li, "Learning deep semantic segmentation network under multiple weakly-supervised constraints for cross-domain remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 20–33, May 2021.

[31] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[32] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890.

[33] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[34] Z. Wang, J. Zhao, R. Zhang, Z. Li, Q. Lin, and X. Wang, "UATNet: U-shape attention-based transformer net for meteorological satellite cloud recognition," *Remote Sens.*, vol. 14, no. 1, p. 104, Dec. 2021.

[35] F. Yang, H. Yang, J. Fu, H. Lu, and B. Guo, "Learning texture transformer network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5791–5800.

[36] Y. Zheng et al., "Semi-supervised adversarial semantic segmentation network using transformer and multiscale convolution for high-resolution remote sensing imagery," *Sensing*, vol. 14, no. 8, p. 1786, 2022.

[37] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 596–608.

[38] D. Berthelot, N. Carlini, I. Goodfellow, N. Papernot, A. Oliver, and C. A. Raffel, "MixMatch: A holistic approach to semi-supervised learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 5049–5059.

[39] D. Berthelot et al., "ReMixMatch: Semi-supervised learning with distribution alignment and augmentation anchoring," 2019, *arXiv:1911.09785*.

[40] T. DeVries and G. W. Taylor, "Improved regularization of convolutional neural networks with cutout," 2017, *arXiv:1708.04552*.

[41] W.-C. Hung, Y.-H. Tsai, Y.-T. Liou, Y.-Y. Lin, and M.-H. Yang, "Adversarial learning for semi-supervised semantic segmentation," 2018, *arXiv:1802.07934*.

[42] L.-C. Chen et al., "Naive-student: Leveraging semi-supervised learning in video sequences for urban scene segmentation," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K. Berlin, Germany: Springer, Aug. 2020, pp. 695–714.

[43] T. Miyato, S.-I. Maeda, M. Koyama, and S. Ishii, "Virtual adversarial training: A regularization method for supervised and semi-supervised learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 8, pp. 1979–1993, Aug. 2019.

[44] X. Chen, Y. Yuan, G. Zeng, and J. Wang, "Semi-supervised semantic segmentation with cross pseudo supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2613–2622.

[45] Y. Ouali, C. Hudelot, and M. Tami, "Semi-supervised semantic segmentation with cross-consistency training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12674–12684.

[46] Z. Ke, D. Qiu, K. Li, Q. Yan, and R. W. H. Lau, "Guided collaborative training for pixel-wise semi-supervised learning," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K. Springer, Aug. 2020, pp. 429–445.

[47] L. Yang, W. Zhuo, L. Qi, Y. Shi, and Y. Gao, "ST++: Make self-trainingWork better for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4268–4277.

[48] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1195–1204.

[49] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "Mixup: Beyond empirical risk minimization," 2017, *arXiv:1710.09412*.

[50] N. Souly, C. Spampinato, and M. Shah, "Semi supervised semantic segmentation using generative adversarial network," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 5688–5696.

[51] L. Cheng, P. Fang, Y. Yan, Y. Lu, and H. Wang, "TRL: Transformer based refinement learning for hybrid-supervised semantic segmentation," *Pattern Recognit. Lett.*, vol. 164, pp. 239–245, Dec. 2022.

[52] J.-X. Wang, S.-B. Chen, C. H. Q. Ding, J. Tang, and B. Luo, "Semi-supervised semantic segmentation of remote sensing images with iterative contrastive network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[53] L. Yang, L. Qi, L. Feng, W. Zhang, and Y. Shi, "Revisiting weak-to-strong consistency in semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7236–7246.

[54] T. Kwak and Y. Kim, "Semi-supervised land cover classification of remote sensing imagery using CycleGAN and EfficientNet," *KSCE J. Civil Eng.*, vol. 27, no. 4, pp. 1760–1773, Apr. 2023.

[55] Z. Zhang, F. Liu, C. Liu, Q. Tian, and H. Qu, "ACTNet: A dual-attention adapter with a CNN-transformer network for the semantic segmentation of remote sensing imagery," *Remote Sens.*, vol. 15, no. 9, p. 2363, Apr. 2023.

[56] T. Xiao, Y. Liu, Y. Huang, M. Li, and G. Yang, "Enhancing multiscale representations with transformer for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605116.

[57] H.-X. Cheng, X.-F. Han, and G.-Q. Xiao, "TransRVNet: LiDAR semantic segmentation with transformer," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 6, pp. 5895–5907, Jun. 2023.

[58] X. Wang et al., "Transformer-based automated segmentation of recycling materials for semantic understanding in construction," *Autom. Construct.*, vol. 154, Oct. 2023, Art. no. 104983.

[59] W. Huang, Y. Shi, Z. Xiong, Q. Wang, and X. X. Zhu, "Semi-supervised bidirectional alignment for remote sensing cross-domain scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 192–203, Jan. 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.

[60] Y. Feng, R. Song, W. Ni, J. Zhu, and X. Wang, "A novel semi-supervised long-tailed learning framework with spatial neighborhood information for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

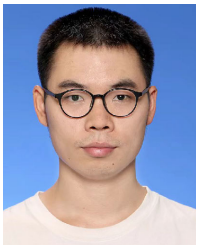
[61] G. Bertò et al., "Classifyber, a robust streamline-based linear classifier for white matter bundle segmentation," *NeuroImage*, vol. 224, Jan. 2021, Art. no. 117402.

[62] C. Zhang, W. Jiang, Y. Zhang, W. Wang, Q. Zhao, and C. Wang, "Transformer and CNN hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408820.

[63] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[64] Y. Liu, Y. Zhang, Y. Wang, and S. Mei, "Rethinking transformers for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617515.

[65] Y. Liu, Y. Zhang, Y. Wang, and S. Mei, "BiTSRS: A bi-decoder transformer segmentor for high-spatial-resolution remote sensing images," *Remote Sens.*, vol. 15, no. 3, p. 840, Feb. 2023.



Yunbo Li received the B.S. degree from the China University of Geosciences (Beijing), Beijing, China, in 2021, where he is currently pursuing the M.S. degree with the School of Land Science and Technology.

His research interests include remote sensing image semantic segmentation and semisupervised learning.



Zhiyu Yi is currently pursuing the Ph.D. degree in surveying and mapping science and technology with the China University of Geosciences (Beijing), Beijing, China.

Her research interests include remote sensing image semantic segmentation and change detection.



Yuebin Wang (Member, IEEE) received the Ph.D. degree from the School of Geography, Beijing Normal University, Beijing, China, in 2016.

He was a Post-Doctoral Researcher at the School of Mathematical Sciences, Beijing Normal University. He is currently an Associate Professor with the School of Land Science and Technology, China University of Geosciences (Beijing), Beijing. His research interests include remote sensing imagery processing and 3-D urban modeling.



Liqiang Zhang (Member, IEEE) received the Ph.D. degree in geoinformatics from the Institute of Remote Sensing Applications, Chinese Academy of Sciences, Beijing, China, in 2004.

He is currently a Professor with the Faculty of Geographical Science, Beijing Normal University, Beijing. His research interests include remote sensing image processing, 3-D urban reconstruction, and spatial object recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:32:45 UTC from IEEE Xplore. Restrictions apply.